

A Computationally Rapid Automated Penalized Likelihood Multiple Changepoint Procedure for Complex Model Structures

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Joint work with Colin Gallagher, Robert Lund, Xueheng Shi

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Come work with me!

- Jeff and Karen Camm
Endowed Professorship in
Data Science and Decision
Intelligence
- Tenure Track Assistant
Professor level
- Start Aug 15 2027
- Submit by 15th Oct 2026
- Open to traditional and
non-traditional backgrounds

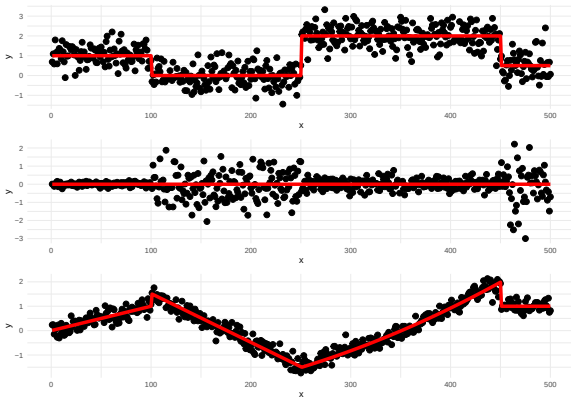


- What are changepoints?
- What do I mean by complex model structures?
- GD-cpt
- Application to climate data

What are changepoints?

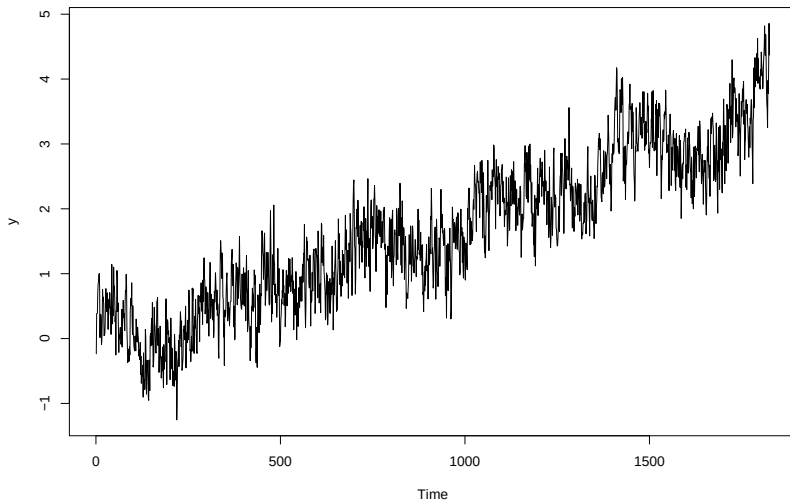
For data $y_1, \dots, y_n$, if a changepoint exists at τ , then $y_1, \dots, y_\tau$ differ from $y_{\tau+1}, \dots, y_n$ in some way.

There are many different types of change.



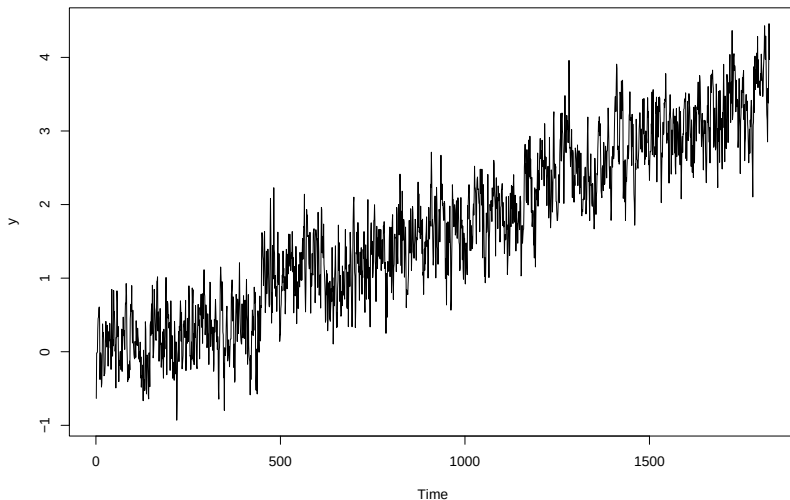
- Model building
- What can change?
- What fitting algorithm to use?
- Penalty selection

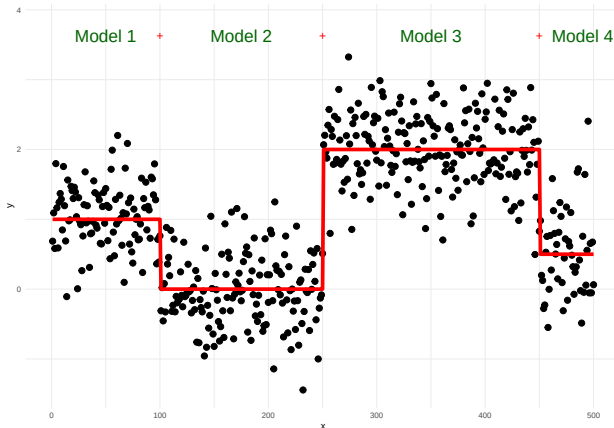
Example data



If you wanted to identify changepoints how would you model this?

Example data





Want to minimize:

$$\min_{m, \tau} \left\{ \sum_{i=1}^{m+1} [\mathcal{C}(y_{\tau_{i-1}:\tau_i})] + \beta m \right\}$$

- PELT: optimizes additive losses in linear computational time
- PDPA: looks at the optimizer as a function of the (single) parameter
- FPOP: combines PELT and PDRA for single parameter settings
- use computational geometry (convex hull) to compute the solution even faster
- factor in constraints (up-down-up, physics constrained)

All of these approaches solve the optimization problem exactly.

How can we identify changes when we have global parameters?

- Allow them to change and fit additive model
 - Not parsimonious
 - Requires post-analysis inference to decide if a change did occur
 - Not suitable for some applications, e.g sub-seasonal changepoints
- Estimate them and remove
 - Robust estimation needed
 - Sensitivity of changepoints to estimates

Ideally we want to fit jointly.

- If we have a good estimate of global parameters, we know how to fit the changepoints
- If we know the changepoints, we can get a good estimate of global parameters

Given initial estimates, we can iterate between these and converge to a solution.

Problem: potential loops

1. Initialise global parameters
2. Fit changepoint model to residuals from global parameters
3. Conditional on the changepoints, update global parameters via a gradient step
4. Refit the changepoint model to the new residuals
5. Iterate 3 and 4 until convergence

Let θ_c changing, and θ_g global parameters respectively.

Theorem (Finite sample convergence to a fixed point)

For $\{Y_t\}$, $t \in 1, \dots, N$, and a changepoint method that guarantees,

$$(\hat{\theta}_c, \hat{\tau}_{1:m}, \hat{m}) = \arg \min_{\tau_{1:m}, m, \theta_c} [C(y_{1:N}, \tau_{1:m}, m, \theta_c | \theta_g) + \beta m]$$

Then,

- 1. GD-cpt converges to a fixed point that depends on the initial $\theta_g^{(0)}$.*
- 2. For the global minimizer, $\tilde{\lambda} = (\tilde{\tau}_{1:\tilde{m}}, \tilde{m}, \tilde{\theta}_c, \tilde{\theta}_g)$, if $\theta_g^{(s)} = \tilde{\theta}_g$ then when the changepoint step is completed, $\theta_c^{(s)} = \tilde{\theta}_c$ and the algorithm stops at this global minimum.*

Theorem (Consistency under infill asymptotics)

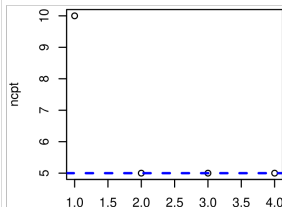
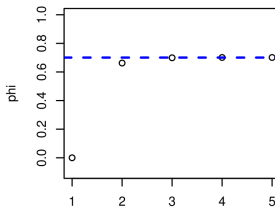
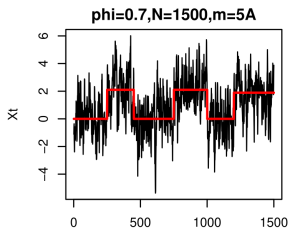
As $N \rightarrow \infty$, when there is asymptotic separation between the changepoint times,

$$\hat{m} \xrightarrow{\mathcal{P}} m, \quad \hat{\tau}_i \xrightarrow{\mathcal{P}} \tau_i \text{ for } 1 \leq i \leq m.$$

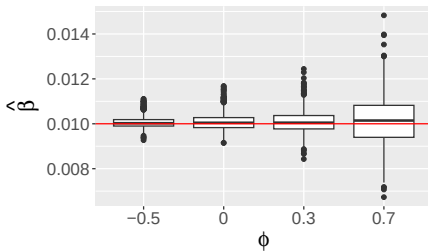
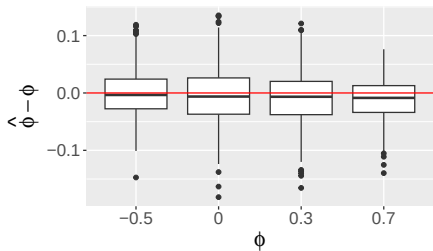
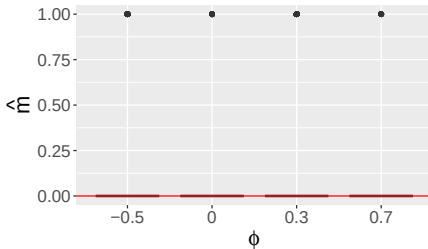
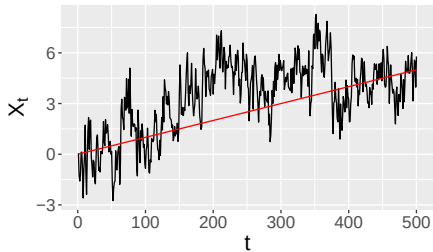
and the global and local parameters are consistently estimated in that

$$(\hat{\beta}_g)_i \xrightarrow{\mathcal{P}} (\beta_g)_i, 1 \leq i \leq d_g; \quad (\hat{\beta}_s)_i \xrightarrow{\mathcal{P}} (\beta_s)_i, 1 \leq i \leq d_s.$$

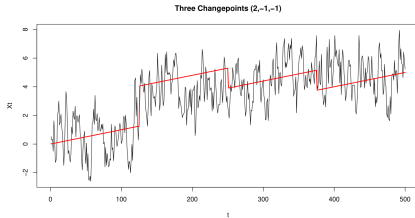
Does it work?



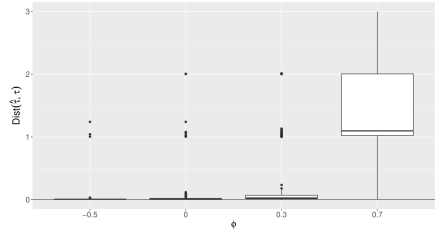
Trend AR1



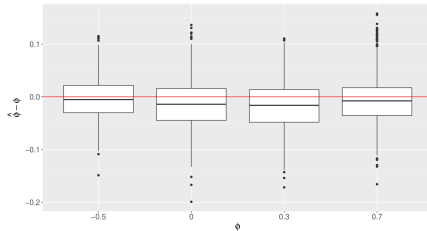
Trend AR1



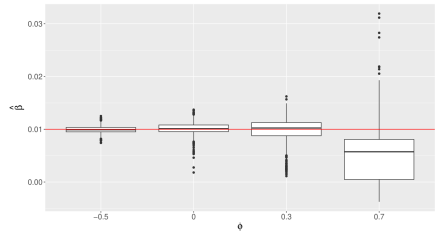
(a) Signal



(b) distance between τ and $\hat{\tau}$

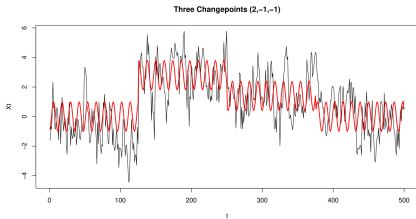


(c) $\hat{\tau} - \tau$

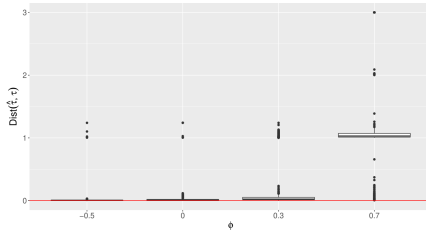


(d) $\hat{\beta} - \beta$

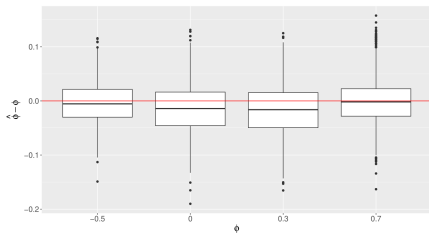
Seasonal AR1



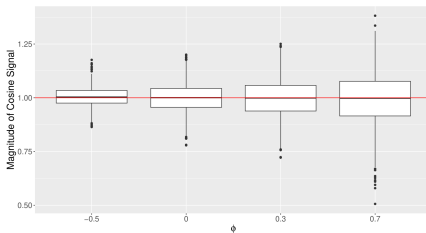
(a) Seasonal signal $T = 12$



(b) distance between τ and $\hat{\tau}$

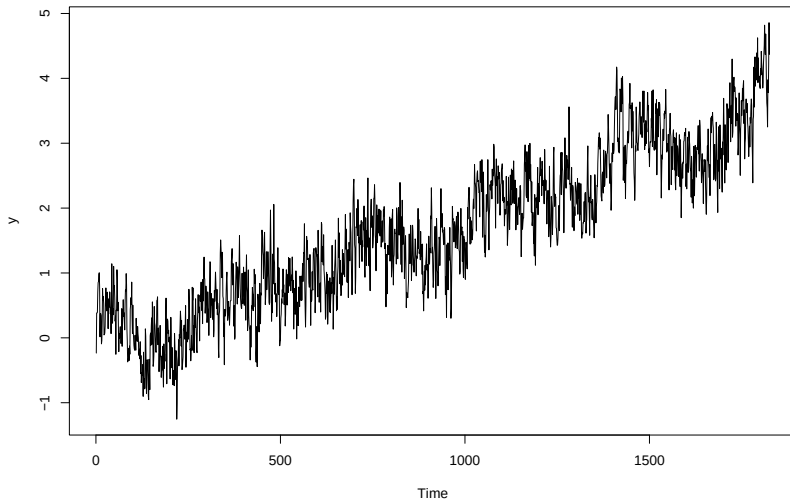


(c) $\hat{\phi} - \phi$

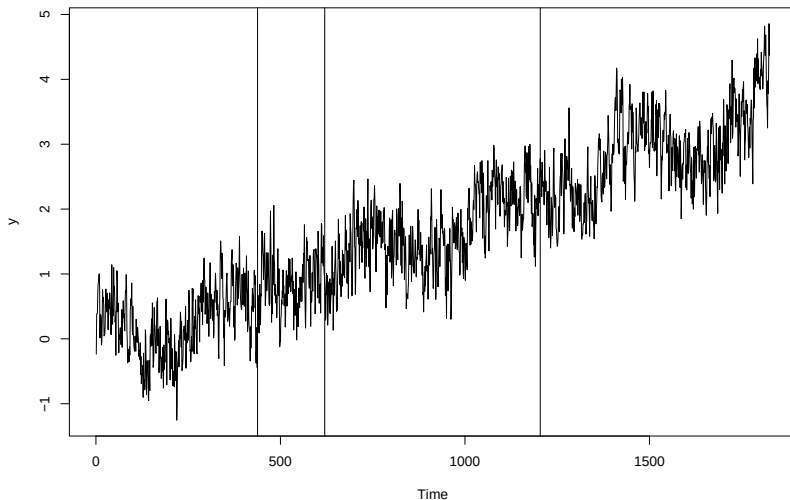


(d) $\hat{A}$

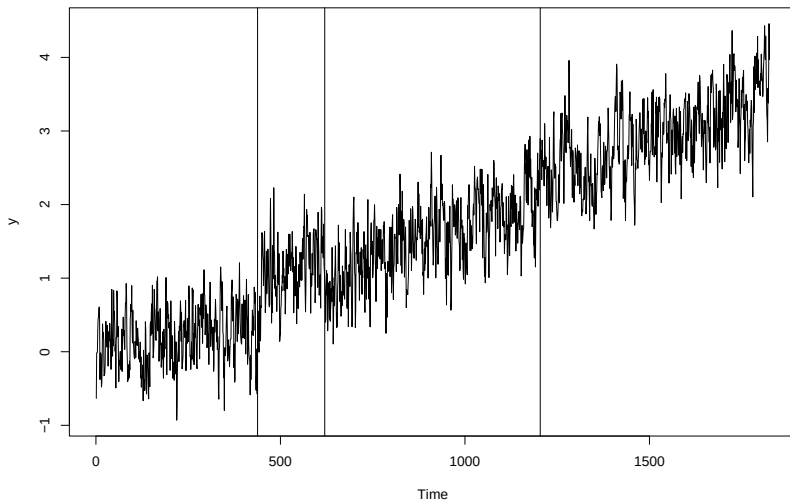
Original Example



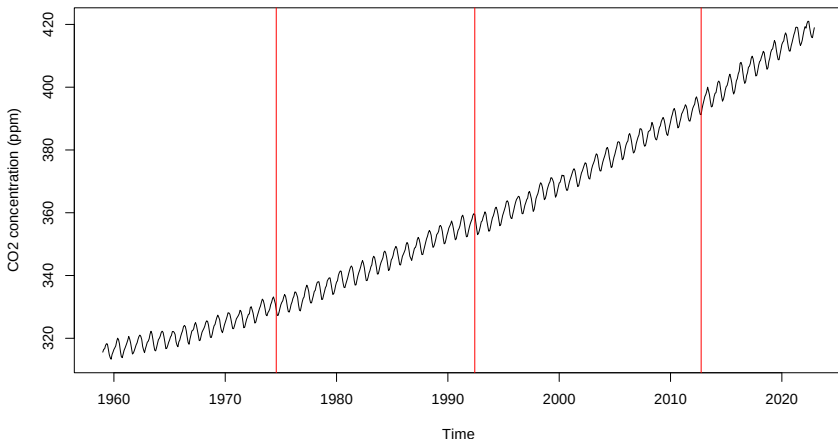
Original Example



Example data



Volcanic eruption (Pinatubo) in 1991, interesting slow down in 2012.



- Changepoints occur in most data
- Detecting and documenting them improves inference
- Non-changing dynamics break additivity that search methods assume
- GD-cpt combines Gradient Descent and any existing search method to identify both local and global parameters efficiently
- This is especially important for data with a fixed seasonal cycle
- I enjoy working with messy real life data . . .
- . . . as often it sparks my next research challenge.